

# Parixit Singh Balot

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## SUMMARY

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B.Tech Computer Science student building agentic AI systems and computer vision pipelines. Fine-tuned vision-language and segmentation models, built an LLM tool-orchestration backend in FastAPI with a React front end, and ship developer tools in Python, JavaScript, and TypeScript. Seeking AI/M, or software engineering internships.

## TECHNICAL SKILLS

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**Languages:** Python, JavaScript, TypeScript, C, SQL

**AI / ML:** PyTorch, Hugging Face Transformers, LoRA fine-tuning, Grounding DINO, PaliGemma, U-Net (segmentation-models-pytorch), computer vision (visual question answering, visual grounding, semantic segmentation, change detection), LLM tool-calling, agentic orchestration, LLM integration (Gemini, Groq, Ollama)

**Backend & Web:** FastAPI, Node.js, REST APIs, React, Vite, HTML/CSS, automation, web scraping, Chrome and VS Code extension development

**Data & Tools:** SQL, database management, Linux, Git and GitHub, pytest, Jupyter, Kaggle, Claude Code

## PROJECTS

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### SatQuery AI: Agentic Vision-Language Assistant for Satellite Imagery Aug 2026

Python, PyTorch, FastAPI, React, TypeScript, Google Earth Engine | [github.com/Parixit007/SAT-AI](https://github.com/Parixit007/SAT-AI)

- Built an agentic assistant where an LLM controller (Gemini or Groq) routes natural-language queries about satellite images to 6 specialist tools and returns evidence-grounded answers with an auditable execution trace.
- Fine-tuned Grounding DINO (DIOR-RSVG, VRSBench) for text-guided grounding, trained a U-Net (ResNet-34) for water-body segmentation, and wrote a LoRA pipeline for PaliGemma-3B VQA, all on Kaggle GPUs.
- Added change detection, optical-SAR fusion, and a groundwater-potential score using an AHP-weighted Google Earth Engine overlay.
- Shipped a FastAPI backend and React + TypeScript + Leaflet interface with 68 pytest tests; the LLM only selects tools, so failures surface as trace warnings instead of crashes.

### WhatsApp AI Bot Mar 2026

Node.js, whatsapp-web.js, Ollama

- Built a personal AI assistant on whatsapp-web.js and locally hosted LLMs via Ollama, with per-contact conversation memory persisted in JSON and multimodal vision support.
- Implemented whitelist-based access control and owner commands issued through self-messaging.

### The Greatest Code Highlighter: VS Code Extension Apr 2026 - Present

TypeScript, VS Code Extension API | [github.com/Parixit007/the-greatest-code-highlighter](https://github.com/Parixit007/the-greatest-code-highlighter)

- Built a VS Code extension that lets developers create persistent code highlights and save them as portable, shareable JSON files that can be committed to Git and reused across environments.
- Used fuzzy matching so highlights stay accurate across code edits, file modifications, session changes, and changes made outside VS Code.

### Poker Bankroll: Self-Hosted LAN Game Tracker Sept 2026

Python (standard library), JavaScript, HTML/CSS | [github.com/Parixit007/poker-bankroll](https://github.com/Parixit007/poker-bankroll)

- Built a zero-dependency Python HTTP server and single-page frontend that lets players join from their phones over Wi-Fi and see live chip counts, betting rounds, pool splits, an undo action, and a full history log.
- Enforced the manager role from the TCP connection's source address and verified player identity with server-issued tokens; wrote a dependency-free QR code encoder for one-scan joining.

### Declutter: Chrome Extension Dec 2025 - Present

JavaScript, Chrome Extension API | [github.com/Parixit007/Declutter](https://github.com/Parixit007/Declutter)

- Built a Chrome extension that closes user-defined tabs by URL from a popup or keyboard shortcut, using a minimal-permission, safety-first design.

## EDUCATION

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### South Asian University, New Delhi Aug 2025 - Jul 2029

B.Tech, Computer Science and Engineering